

Multi-Stage Planning of Distribution Network With High Penetration Renewable Energy Considering Reliability Index

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Abstract—In this study, a multi-stage planning model of the distribution network with high photovoltaic (PV) penetration considering operational reliability is presented. The operation reliability cost consisting of three modified reliability indices is imposed in the objective function in the proposed distribution systems planning model. Additionally, the distribution network reconfiguration including microgrid is considered at the operation level to improve system operation reliability. Nonlinear expressions of modified reliability indices aggravate the complexity of the proposed planning model. To address the computational burden, the planning model is simplified as a mixed integer linear programming (MILP) model, which have an efficient computational behavior with exiting MILP solvers. The benefits of imposing operation reliability cost in the objective function and network reconfiguration including microgrid as operation strategies, are discussed. The effectiveness of the proposed model is tested by numerical results.

Index Terms—Distribution network, multi-stage planning, reliability index, photovoltaic, mixed integer linear programming.

I. INTRODUCTION

THE objective of distribution network planning is to determine the optimal size, location and time for the addition and replacement of energy generators, transmission lines and distribution feeders by minimizing investment and operation costs over a planning horizon [1], [2], [3]. With the aim of facilitating the coordination and operation safety and reducing the short-circuit current of distribution network, the radial topology

is the main structure of distribution system. Hence, the radiality constraint is considered in almost all of the distribution network planning models [4], [5], [6].

There is an increasing integration of renewable energy, such as photovoltaic (PV) into the distribution system due to the depletion of traditional fossil fuels and the rise of environmental safety. The rapid growth in renewable energy into distribution systems has introduced several challenges to secure and stable grid operation. The distribution network planning considering the integration of renewable energy has received additional attention among decision-makers [7], [8], [9], [10]. In [11], a multi-agent framework is proposed to address the expansion planning problem in a restructured active distribution network, where the objective and techno-economic constraints of participating agents are addressed in the expansion planning of power network and distributed energy. In [12], to achieve high operation reliability for the distribution network, a novel multi-stage expansion planning model for mesh distribution networks is proposed, in which reliability assessment is explicitly implemented as constraints. In [13], a novel bi-level coordinated planning model which combines the short-time-scale operation problem with the long-time-scale planning problem is proposed.

Although the system will mostly operate in the normal condition, the power system reliability has become an inevitable factor in power system studies due to electricity power plays an important role in human society that even small interruptions cannot be tolerated. Hence, power system reliability has become an inevitable factor in power system studies. More specifically, since majority of system instability are originated from grid fault, reliability considerations in this system have been being of great interest [14], [15], [16]. Importance of reliability considerations has also been reflected in many articles, and the reliability index is generally utilized in research publications to reflect system operation reliability level. In [17], a multi-objective multi-stage model for distribution network expansion planning is proposed, where the system operation reliability is considered by incorporating not-served energy cost in the objective function. In [18], a novel framework is proposed for planning a distribution network, where reliability indices is added in the planning framework to reflect system automation capabilities. In [19], all the expected energy not served index

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(EENS), system average interruption duration index (SAIDI), and average energy not supplied (AENS) indices are taken into account to form a MILP problem to improve system operation reliability.

The review of references reveals that the reliability indexes utilized in these articles mainly focus on the demand of power consumers [20], [21], [22], [23], [24], the literature lacks an effective assessment model considering the demand of PV owners to reflect the operation reliability of the system with high penetration level renewable energy. In the past, the demand of PV owners was not considered to assess the system operation reliability in the planning strategy, given how few systems were deployed. However, as more PV generation systems are installed, and inverters become more advanced, PV plays a more and more important role in power supply. The PV is generally constructed and owned by third parties due to corresponding policies in many countries, and the PV owner trades PV power with the distribution system operator. Frequent PV output power interruption due to outage of grid lines will lead to huge revenue loss for PV owners, which may discourage PV producers from installing PV. Hence, the reliability index considering all the operation demands for consumers and PV owners should be incorporated into the planning model to evaluate the operation reliability of the distribution network. Additionally, few distribution network planning strategies has considered the impact of high penetration renewable energy on distribution operation reliability with the occurrence of the transmission line fault. Hence, a novel multi-stage planning model is proposed in this article to improve the operation reliability of the distribution network with high penetration PV. The main contributions of this article are as follows:

- 1) A multi-stage planning strategy is proposed to improve the operation reliability of the distribution network, in which the infrastructure investment options and operation strategies of each planning stage are co-optimized by minimizing investment cost, operation cost and operation reliability cost in the whole planning horizon.
- 2) A novel operation reliability cost formulation consisting of three modified reliability indices (modified EENS, modified SAIDI, and modified SAIFI) is developed, where the revenue lost associated with the undelivered PV output power due to the outage of branch as well as the cost incurred by the branch power flow based penalty scheme are considered to evaluate the operation reliability of the system with high penetration PV.
- 3) A mixed integer linear programming (MILP) model that guarantees the optimal global solution of the planning problem is developed, which have an efficient computational behavior with exiting MILP solvers.

II. IMPACT OF HIGH PV PENETRATION ON DISTRIBUTION NETWORK OPERATION RELIABILITY

PV generation has been widely deployed at the distribution network due to the clean and green energy it provides. The connected point and capacity of PV generation is mainly

determined by the third parties in the whole planning stage without considering the system line capacity and load demand distribution. Additionally, the Maximum Power Point Tracking (MPPT) control strategy is generally utilized in the PV controller to maximize PV output [25], [26], which will not be affected by system structure and load demand. Hence, the high penetration PV will have huge impact on system power flow, which is shown in Fig. 1.

Fig. 1 depicts the power flow performance in the case of line fault in the traditional distribution network and high PV penetration distribution system, respectively. In the traditional distribution network, the load demand is supported by the upper grid, the failure of the downstream distribution line result in no influence on the upstream customer. Given the occurrence of a line to ground fault in a high penetration PV system, the distribution line will be removed, and the output power generated by PV will be transmitted through other lines, thus leading to line overload and system over-voltage. Additionally, the active power imbalance between the PV generation and customer demand due to line fault will result in a frequency drop, thus leading to system frequency instability.

It can be concluded that high PV generation has a negative impact on system operational reliability, which will be higher for high penetration levels. Such an emerging issue in high PV penetration distribution networks highlights the importance of considering operational reliability in the planning model. The investment cost and stable operation cost are generally considered as optimal function in conventional planning models, which may reduce system operational reliability. With the aim of reducing the impact of PV on system operational reliability and improving the stability of the planning distribution network, the operation reliability cost comprised of modified reliability indices is imposed in the proposed planning model, and the distribution network reconfiguration including microgrid is considered, the detail of the planning model will be explained in the next section.

III. PROBLEM DESCRIPTION

The proposed investment and operation strategy in the co-optimization planning model of the distribution network with high penetration renewable energy is depicted in Fig. 2. It can be seen from Fig. 2 that the dimension of the planning stage in the proposed planning model is set as a function of η , the dimension of the time stage in each planning stage is set as a function of t , and the investment decision in various time stage is deciding on the optimization result of the building plan. Additionally, the dimension of renewable energy scenarios is depicted as a function of s , which differ from one scenario to the next to deal with the uncertainties of PV generation. In the operation strategy, the network topology and power generation of DG and SUB are optimized at the same time based on the load and PV data in each scenario to minimize the stable operation cost and transient operation cost.

The assumption of PV in the distribution network is as follows:

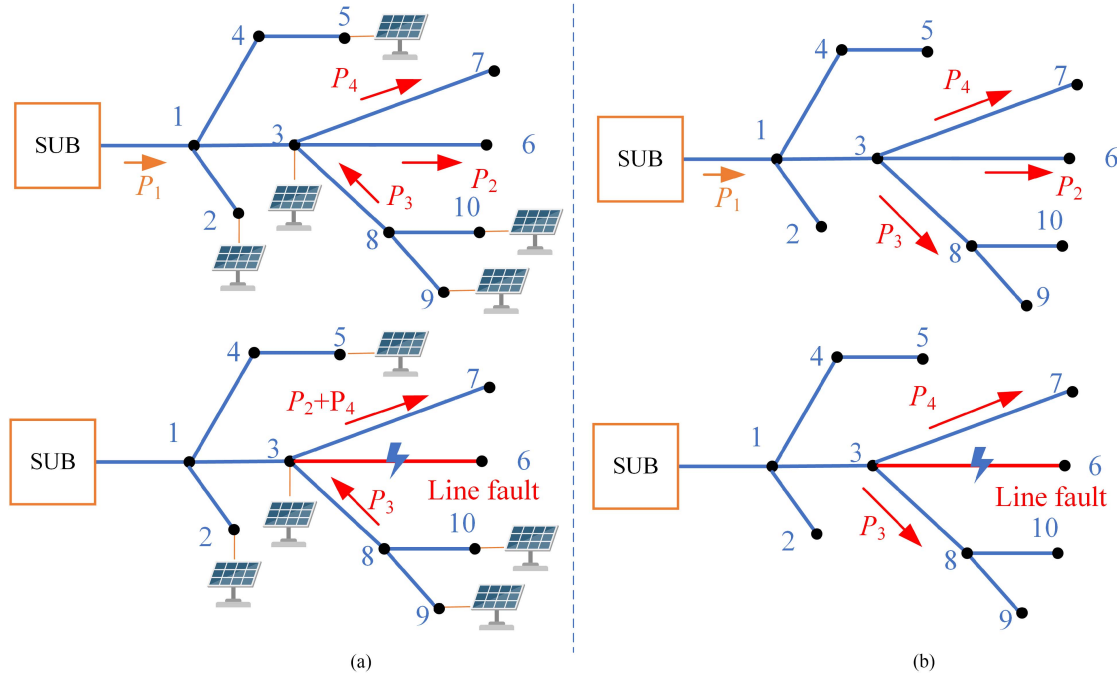


Fig. 1. Power flow performance in the case of line fault in the traditional distribution network and high PV penetration distribution system. (a) High penetration PV distribution system, (b) Traditional distribution network.

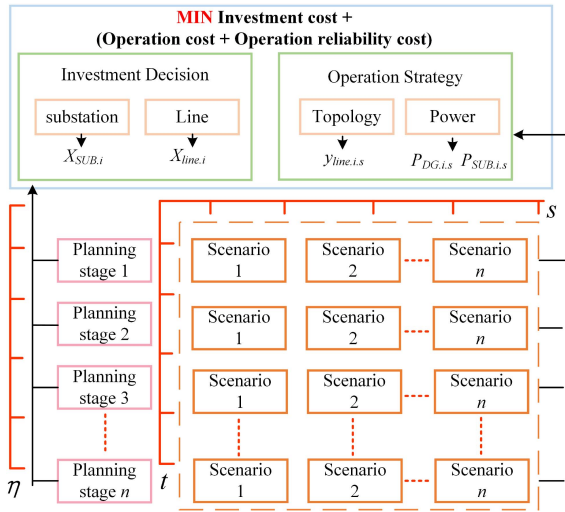


Fig. 2. Proposed co-optimization planning strategy.

- 1) The DG deployed at the distribution level is constructed and owned by the distribution network operator, while the PV is added and owned by the third parties (independent PV producers).
- 2) The PV owner has a power trade contract with the power grid operator. All the power generated by PV is accepted by the distribution network operator at a fixed price.

A. Objective Function

In this article, a multi-stage planning model considering system operation reliability level is presented. The objective

function of the proposed model to be minimized is comprised of investment cost ($C_{inv.t,\eta}$), operation cost ($SC_{op.s,t,\eta}$), and operation reliability cost ($TC_{op.s,t,\eta}$), which are as follows:

$$\min \sum_{\eta} \sum_t \frac{1}{(1+\gamma)^{((\gamma-1)T+t)}} C_{inv.t,\eta} + \sum_{\eta} \sum_t \frac{1}{(1+\gamma)^{((\gamma-1)T+t)}} \sum_{s \in \Omega_s} (SC_{op.s,t,\eta} + TC_{op.s,t,\eta}) \quad (1)$$

where γ is the discount rate of interest, T is the optimization time in each planning stage, t is the index for time stage of planning stage, η is the index for planning stage, Ω_s is the set of load and PV scenario.

• Investment Cost

The investment cost at time stage t of planning stage η , $C_{inv.t,\eta}$, including: 1) the cost of adding new branches, 2) the cost of replacing existing branches, 3) the cost of adding new substation, 4) the cost of adding new distributed generator.

$$C_{inv.t,\eta} = \sum_{i \in AL} \sum_{j \in CAL_k} C_{AX}^j X_{A.l,i,t,\eta}^j + \sum_{i \in RL} \sum_{j \in CRL_k} C_{RX}^j X_{R.l,i,t,\eta}^j + \sum_{i \in ASub} C_{Sub} X_{A.Sub,i,t,\eta} Q_{Sub,i} + \sum_{i \in ADG} \sum_{j \in CDG_k} C_{ADG} X_{A.DG,i,t,\eta}^j Q_{DG,i} \quad (2)$$

where AL is the set of lines to be added, CAL is the set of candidate choices of adding a new line, C_A is cost of adding new lines, $X_{A.l.i.t.\eta}^j$ is a binary variable associated with the infrastructure investment decision, which is 1 when the line i is added using candidate choice j , RL is the set of lines to be replaced, CRL is the set of candidate choices of replacing a line, C_R is cost of replacing a line, $X_{R.l.i.t.\eta}^j$ is a binary variable, which is 1 when the line i is replaced using candidate choice j , C_{Sub} is cost of replacing a line, $X_{A.Sub.i.t.\eta}$ is a binary variable, which is 1 when the substation i is added, $Q_{Sub.i}$ is the installed capacity of substation, ADG is the set of node to build up DG, $CADG$ is the set of candidate choices of building up a DG, C_{DG} is cost of building up a DG, $X_{DG.l.i.t.\eta}^j$ is a binary variable, which is 1 when a DG is added in node i using candidate choice j , $Q_{Sub.i}$ is the installed capacity of DG.

- Operation Cost

The operation cost at time stage t of planning stage η , $SC_{op.t.\eta}$, including: 1) the cost of power purchase from the upper power grid, 2) the cost of DG operation, 3) the maintenance cost of DGs, 4) the maintenance cost of power lines.

$$SC_{op.s.t.\eta} = C_{l.s.t.\eta} + C_{DG.s.t.\eta} + C_{Sub.s.t.\eta} \quad (3)$$

$$C_{l.s.t.\eta} = \sum_{i \in EL} O_E Y_{E.i.t.\eta} + \sum_{i \in AL} \sum_{j \in CAL_k} O_A Y_{A.i.t.\eta}^j + \sum_{k \in RL} \left(O_0 Y_{0.k.s.t.\eta} + \sum_{j \in CEL_k} O_R Y_{R.i.t.\eta}^k \right) \quad (4)$$

$$C_{DG.s.t.\eta} = \sum_{EDG} \left(O_{DG} P_{DG.i.s.t.\eta} + O_{DG.fix} Y_{E.DG.i.s.t.\eta}^j \right) + \sum_{ADG} \left(O_{DG} P_{DG.i.s.t.\eta} + O_{DG.fix} Y_{A.DG.i.s.t.\eta}^j \right) \quad (5)$$

$$C_{Sub.s.t.\eta} = \sum_{i \in ESub} O_{Sub} P_{Sub.i.s.t.\eta} + \sum_{ASub} O_{Sub} Y_{ASub.i.s.t.\eta} P_{Sub.i.s.t.\eta} \quad (6)$$

where s is the index for load and PV scenario, $C_{l.s.t.\eta}$ is the operation cost of lines, $C_{DG.s.t.\eta}$ is the operation cost of DGs, $C_{Sub.s.t.\eta}$ is the operation cost of substation, O_R , O_A and O_E are the maintenance cost of replaced line, added line and existing line, respectively. $Y_{E.l}$, $Y_{A.l}$ and $Y_{R.l}$ are the binary variable associated with validation flag of existing lines, added lines and replaced lines respectively, which are 1 if associated line is in-use. O_{DG} and $O_{DG.fix}$ are the generating cost and maintenance cost of DG, respectively. $P_{DG.i.s.t.\eta}$ is the generating power of DG, $Y_{E.DG.i.s.t.\eta}$ and $Y_{A.DG.i.s.t.\eta}$ are the binary variable, which are 1 if associated DG is in-use. O_{Sub} is the cost of buying electricity from substation, $P_{Sub.i.s.t.\eta}$ is the purchased power from the substation i , and $Y_{A.Sub.i.s.t.\eta}$ is the binary variable, which are 1 if substation i is in-use.

B. Traditional Distribution Network Constraints

Traditional distribution network constraints contained in the proposed multi-stage planning model are divided into categories as follows:

- 1) Power Balance and Power Limit Constraints
- 2) Voltage Balance and Voltage Limit Constraints.
- 3) Construction Logic Constraints

The detail of traditional distribution network constraints, including formulation, decision variables, and the validation of these constraints, can be seen from [6].

C. Proposed Operation Reliability Cost

As mentioned before, the operation reliability of distribution networks has attracted more and more attention in distribution network studies with the rapid growth of renewable energy, and the importance of system operation reliability has been reflected in many research. However, the reliability index considered in these planning models, such as EENS, SAIDI, and SAIFI, is mainly focused on the operation reliability demand of the consumer. Nowadays, PV plays such an accentuated role in power supply with the rapid development of corresponding technology. The PV is generally constructed and owned by third parties due to corresponding policies in many countries, and the PV owner trades PV power with the distribution system operator. With the occurrence of line fault, the revenue lost for PV owners due to power interruption may discourage PV producers from installing PV, and the impact of PV on power flow performance may deteriorate system operation stability (described in Section II).

Motivated by these points, an operation reliability cost based on three modified reliability index (modified EENS, modified SAIDI, modified SAIFI) is proposed for the planning model of the distribution network with high penetration PV, which comprise two terms:

- 1) Penalty cost incurred of undelivered energy and PV power loss.
- 2) Penalty cost incurred of the negative impact of PV on system operation reliability.

The equation is shown in (7), and the undelivered energy and PV power loss can be calculated using (9)–(11).

$$TC_{t.\eta} = PC_{loss.t.\eta} + PC_{flow.t.\eta} \quad (7)$$

$$PC_{loss.t.\eta} = C_{EENS} MEENS_{t.\eta} + C_{SAIFI} MSAIFI_{t.\eta} + C_{SAIDI} MSAIDI_{t.\eta} \quad (8)$$

$$MEENS_{t.\eta} = \sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{load}} \lambda_i \delta_i P_{load.i.s.t.\eta}^{loss} \frac{Dur_{load.i.s.t.\eta}}{365 * 24} + \sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{PV}} \lambda_i \delta_i P_{PV.i.s.t.\eta}^{loss} \frac{Dur_{PV.i.s.t.\eta}}{365 * 24} \quad (9)$$

$$MSAIFI_{t.\eta} = \frac{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{load}} \lambda_i N_{load.i.s.t.\eta}}{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{load}} N_{load.i.s.t.\eta}}$$

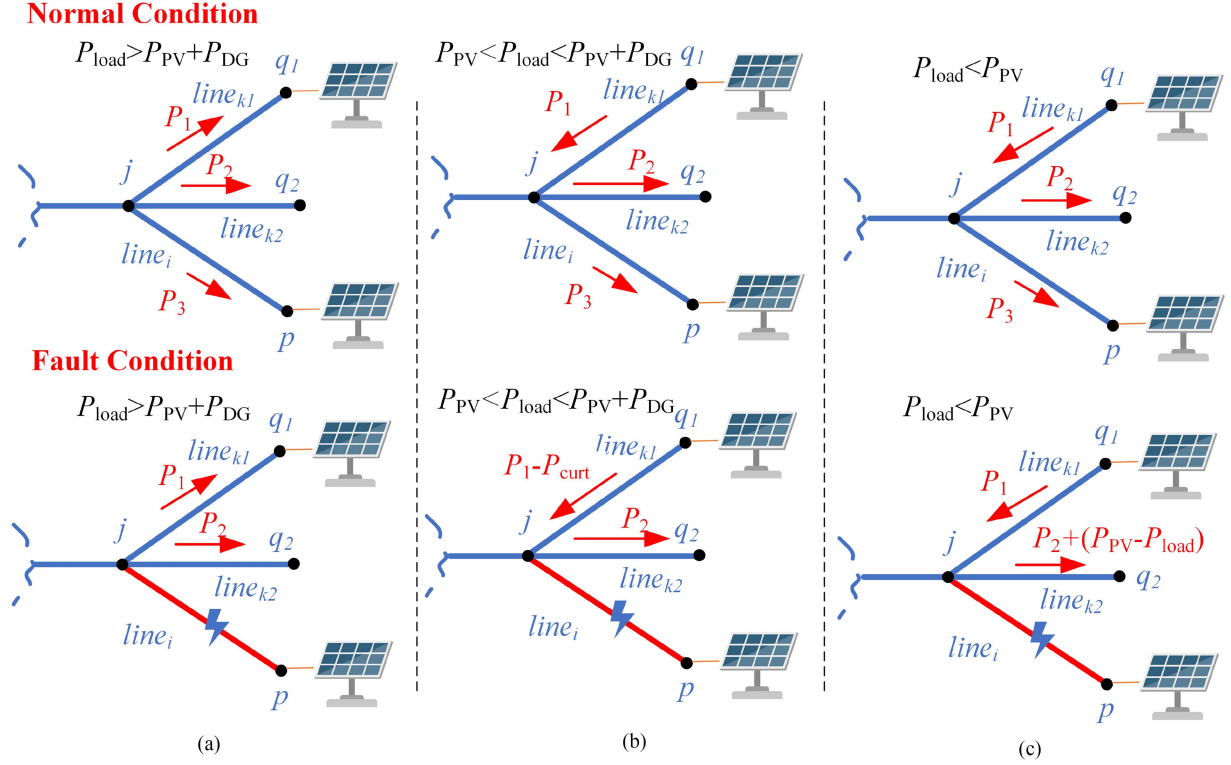


Fig. 3. Power flow under normal and fault conditions.

$$+ \frac{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{PV}} \lambda_i N_{PV.i.s.t.\eta}}{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{PV}} N_{PV.i.s.t.\eta}} \quad (10)$$

$$MSAIDI_{t,\eta} = \frac{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{load}} \lambda_i \delta_i N_{load.i.s.t.\eta}}{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{load}} N_{load.i.s.t.\eta}} + \frac{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{PV}} \lambda_i \delta_i N_{PV.i.s.t.\eta}}{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{PV}} N_{PV.i.s.t.\eta}} \quad (11)$$

where $PC_{loss.t,\eta}$ is the penalty cost incurred of undelivered energy and PV power loss. $PC_{flow.t,\eta}$ is the penalty cost incurred of the negative impact of PV on system operation reliability. C_{EENS} , C_{SAIFI} and C_{SAIDI} are the cost factor of MEENS, MSAIFI and MSAIDI, respectively. λ_i and δ_i are the average number and duration of yearly interruption of load node. $P_{load.i.s.t.\eta}^{loss}$ is the curtailed load demand, $P_{PV.i.s.t.\eta}^{loss}$ is the curtailed PV output power. $Dur_{load.i.s.t.\eta}$ is the load demand duration, $Dur_{PV.i.s.t.\eta}$ is the PV output power duration. $N_{load.i.s.t.\eta}$ is the number of customers at load node i , $N_{PV.i.s.t.\eta}$ is the number of PV at load node i . Ω_{PV} is the set of load node with PV.

In Fig. 3, various power flows under normal and fault conditions are depicted to analyze the impact of PV on system operation reliability with the occurrence of a line fault. Under normal conditions with the power flow in $line_{k1}$ of $P_1 > 0$ due to the load demand P_{load} in node q_1 is greater than the sum of P_{PV} and P_{DG} , the direction of system power flows under the fault condition is unchanged, which has no bad influence of system stability [see Fig. 3(a)]. With the power flow in $line_{k1}$ of $P_1 < 0$, and the load demand P_{load} in node q_1 is greater

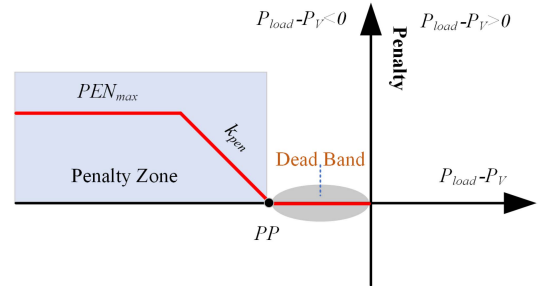


Fig. 4. General graph of penalty scheme.

than the P_{PV} but lower than the sum of P_{PV} and P_{DG} , the negative power flow will have no influence on other adjacent lines under the fault condition according to adjust the output power of DGs [see Fig. 3(b)]. For the normal condition that all the power flow $P_1 < 0$ and the load demand P_{load} in node q_1 is lower than P_{PV} , the generated power by uncontrollable resource PV will flows from node q_1 to q_2 through branch $line_{k2}$ [see Fig. 3(c)], which may lead to mal-operation of system protection and deteriorate system operation reliability.

Based on the above analysis, a penalty method is proposed to calculate the penalty cost incurred of the impact of PV on system stability, where assumed that the distribution network operator is under a penalty regime [27], [28]. In this regime, the distribution network operator will be penalized if they are unable to provide an acceptable reliability plan to limit the negative impact of PV on system stability under the fault condition. The general shape of the proposed penalty scheme is illustrated in Fig. 4. There is

a dead band zone with no penalty around the benchmark value, which implies the acceptable value of negative branch power flow, where no penalty exists.

$$PC_{flow.t.\eta} = \sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{nbr}} \sum_{k \in \Omega_{alt}} \lambda_{l.i.k} \delta_{l.i.k} \frac{Dur_{l.i.s.t.\eta}}{365 * 24} PC_{flow.l.i.s.t.\eta}^{jp} \quad (12)$$

$$PC_{flow.l.i.s.t.\eta}^{jp} = \sum_{k \in \Omega_{nbr}} PEN_{l.k.s.t.\eta}^{jq} \quad (13)$$

$$P_{l.k.s.t.\eta}^{dir} = P_{l.k.s.t.\eta}^{load} - P_{load.k.s.t.\eta}^{PV} \quad (14)$$

$$PEN_{l.k.s.t.\eta}^{jp} = \max \{0, \min \{ (PP - P_{l.k.s.t.\eta}^{dir}) k_{pen}, PEN_{max} \} \} \quad (15)$$

where $\lambda_{l.i.k}$ and $\delta_{l.i.k}$ are the failure rate and repair time of line i using candidate choice k , Ω_{nbr} is the set of network branch, Ω_{alt} is the set of candidate choice for branch i , $Dur_{l.i.s.t.\eta}$ is the branch power flow duration. $PC_{flow.l.i.s.t.\eta}^{jp}$ is the penalty cost incurred of the failure of line i , where the head and end node are j and p , respectively. $P_{l.i.s.t.\eta}^{dir}$ is a auxiliary variables associated with the difference between load demand and PV output power of branch k . $P_{l.k.s.t.\eta}^{PV}$ is the sum of PV output power of branch k , $P_{l.k.s.t.\eta}^{load}$ is the sum of load demand of branch k , $PEN_{l.k.s.t.\eta}$ is the penalty cost for the line fault on k -th branch. PP is the penalty point, PEN_{max} is the penalty caps, k_{pen} is the incentive rates of penalty zones.

D. Proposed Topology Logic Constraints

The radial structure is used as a set of topology logic constraints in the traditional planning model, given most of the distribution networks are operated radially. However, as mentioned before, the radial structure may have a bad impact on system operation reliability in the whole planning horizon with the continuous growth of PV installation capacity. In order to improve the operational reliability of the system with high penetration renewable energy, the topology logic constraints for flexible system structure is proposed in this study to restrict the system operating conditions, where all the radial structure and microgrid are allowed in the whole planning stage:

Condition 1: Two substations are not allowed to be connected directly.

Condition 2: All the substations in the distribution network and its connected load nodes need to form a radial network.

Condition 3: Separated load node with DGs is allowed if microgrid operation is allowed in the operation of system.

These constraints play important roles in the reduction of the planning search area in the algorithm, and can also eliminate unreasonable planning results.

$$\sum_{j \in CAL_i} Y_{A.l.i.s.t.\eta}^j = f_{A.l.i.s.t.\eta}^{pq+} + f_{A.l.i.s.t.\eta}^{qk-} \quad (16)$$

$$Y_{E.l.i.s.t.\eta}^j = f_{E.l.i.s.t.\eta}^{pq+} + f_{E.l.i.s.t.\eta}^{qk-} \quad (17)$$

$$Y_{0.l.k.s.t.\eta} + \sum_{j \in CRL_i} Y_{R.l.i.s.t.\eta}^j = f_{R.l.i.s.t.\eta}^{pq+} + f_{R.l.i.s.t.\eta}^{qk-} \quad (18)$$

$$\sum_{i \in \Omega_l} f_{A.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{qk-} = 0 \quad \forall q \in \Omega_{Sub} \quad (19)$$

$$\sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{qk-} = 0 \quad \forall q \in \Omega_{Sub} \quad (20)$$

$$\sum_{i \in \Omega_l} f_{R.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{R.l.i.s.t.\eta}^{qk-} = 0 \quad \forall q \in \Omega_{Sub} \quad (21)$$

$$\sum_{i \in \Omega_l} f_{A.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{qk-} = 1 \quad \forall q \in \Omega_{load} \quad (22)$$

$$\sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{qk-} = 1 \quad \forall q \in \Omega_{load} \quad (23)$$

$$\sum_{i \in \Omega_l} f_{R.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{R.l.i.s.t.\eta}^{qk-} = 1 \quad \forall q \in \Omega_{load} \quad (24)$$

$$f_{A.l.i.s.t.\eta}^{pq+} + f_{A.l.i.s.t.\eta}^{qk-} \leq \gamma_q \quad \forall q \in \Omega_{tr} \quad (25)$$

$$f_{R.l.i.s.t.\eta}^{pq+} + f_{R.l.i.s.t.\eta}^{qk-} \leq \gamma_q \quad \forall q \in \Omega_{tr} \quad (26)$$

$$f_{E.l.i.s.t.\eta}^{pq+} + f_{E.l.i.s.t.\eta}^{qk-} \leq \gamma_q \quad \forall q \in \Omega_{tr} \quad (27)$$

$$\begin{aligned} & \left(\sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{qk-} \right) \\ & + \left(\sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{qk-} \right) \\ & + \left(\sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{qk-} \right) geq 2\gamma_q \quad \forall q \in \Omega_{Sub} \end{aligned} \quad (28)$$

$$\sum_{i \in \Omega_l} f_{A.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{A.l.i.s.t.\eta}^{qk-} = 1 - \xi_{q.s.t.\eta} \quad \forall q \in \Omega_{DG} \quad (29)$$

$$\sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{qk-} = 1 - \xi_{q.s.t.\eta} \quad \forall q \in \Omega_{DG} \quad (30)$$

$$\sum_{i \in \Omega_l} f_{R.l.i.s.t.\eta}^{pq+} + \sum_{i \in \Omega_l} f_{R.l.i.s.t.\eta}^{qk-} = 1 - \xi_{q.s.t.\eta} \quad \forall q \in \Omega_{DG} \quad (31)$$

$$\sum_{i \in \Omega_l} f_{R.l.i.s.t.\eta}^{pq-} + \sum_{i \in \Omega_l} f_{R.l.i.s.t.\eta}^{qk+} \leq M(1 - \xi_{q.s.t.\eta}) \quad \forall q \in \Omega_{DG} \quad (32)$$

$$\sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{pq-} + \sum_{i \in \Omega_l} f_{E.l.i.s.t.\eta}^{qk+} \leq M(1 - \xi_{q.s.t.\eta}) \quad \forall q \in \Omega_{DG} \quad (33)$$

$$\sum_{i \in \Omega_l} f_{A.l.i.s.t.\eta}^{pq-} + \sum_{i \in \Omega_l} f_{A.l.i.s.t.\eta}^{qk+} \leq M(1 - \xi_{q.s.t.\eta}) \quad \forall q \in \Omega_{DG}$$

$$\leq M(1 - \xi_{q.s.t.\eta}) \quad \forall q \in \Omega_{DG} \quad (34)$$

$$\xi_{q.s.t.\eta} \leq \sum_{i \in CADG_j} Y_{DG.i.s.t.\eta}^j \quad (35)$$

$$\begin{aligned} & \sum_{ij \in \Omega_l} \left(f_{A.l.i.s.t.\eta}^{ij+} + f_{A.l.i.s.t.\eta}^{ij-} \right) \\ & + \sum_{ij \in \Omega_l} \left(f_{R.l.i.s.t.\eta}^{ij+} + f_{R.l.i.s.t.\eta}^{ij-} \right) \\ & + \sum_{ij \in \Omega_l} \left(f_{E.l.i.s.t.\eta}^{ij+} + f_{E.l.i.s.t.\eta}^{ij-} \right) \\ & = N_{node} - \sum_{i \in \Omega_{tr}} (1 - \gamma_q) - \sum_{q \in \Omega_{DG}} \xi_q - \sum_{i \in \Omega_{Sub}} Y_{Sub.i.s.t.\eta} \end{aligned} \quad (36)$$

where $f_{A.l.i.s.t.\eta}^{pq+}$, $f_{E.l.i.s.t.\eta}^{pq+}$ and $f_{R.l.i.s.t.\eta}^{pq+}$ are the auxiliary binary variable associated with the direction of existing lines, lines to be added, lines to be replaced respectively, which are 1 if node p is parent node of node q . $f_{l.i.s.t.\eta}^{qk-}$ is auxiliary binary variable, which are 1 if node k is parent node of node q . Ω_{Sub} is the set of substation nodes, Ω_{load} is the set of load nodes without any candidate connection nodes for DGs, Ω_{tr} is the set of transfer nodes, Ω_{DG} is the set of load nodes with candidate connection nodes for DGs. γ_q is the auxiliary binary variable for transfer node, which are 1 if the transfer node q is connected. ξ_q is the auxiliary binary variable associated with microgrid operation in the operation stage, which are 1 if the node q is allowed to be an independent node. N_{node} is the total number of nodes.

Validation of the proposed topology logic constraints: As mentioned before, three conditions need to be satisfied in the proposed planning model. **Condition 1** and **Condition 2** are satisfied by (16)–(28). Equation (16)–(25) are the radiality constraints, which ensure that the distribution network corresponds to a radial structure connected to the main substation by adding the auxiliary variable, f_l^{pq+} , f_l^{pq-} corresponding to each branch. In the spanning tree constraints, all the substation nodes have no parent node, and all the load nodes without any candidate DGs connection point only have one parent node. Equation (26)–(28) are the generalization of radiality constraints with transfer nodes which refer to simply one node with no DG, PV, or demand. They are applied in this planning model to interconnect load nodes. **Condition 3** is satisfied by (29)–(36), which are the microgrid operation constraints. Equation (29)–(31) indicates that the load node with candidate DGs connection point has no parent node when they operate at microgrid mode. Otherwise, they have only one parent node. Equation (32)–(35) indicates that there is no child node for the load node with candidate DGs connection point operating at microgrid mode, or these inequalities will be relaxed by constant M . Equation (36) shows that the load node with candidate DGs connection point is not allowed to be separated load before the relevant DGs are applied.

Hence, the feasible solution of the proposed multi-stage planning problem satisfies both **Condition 1**, **Condition 2**, and **Condition 3** according to these constraints.

IV. LINEARIZATION

Note that the proposed multi-stage planning model is a mixed-integer non-linear programming (MINLP) model, which is hard to be solved by the commercial solver. the modified reliability index in (9)–(11) and the penalty cost in (12) are non-linear expressions, which will be linearized in this section with the aim of obtaining MILP formulation for the planning problem.

• Linearization of MEENS

It can be seen from (9) that the main issue regarding the calculation of MEENS is to calculate λ_i and δ_i given both of these variables are associated with the distribution network topology, which is unknown during the whole planning horizon. Some assumptions have been made in this section to address this issue, which is as follows:

- 1) All the lines added to the distribution network have a protective device, such as a fuse and breaker, to disconnect the fault line and corresponding downstream network in the case of a line fault.
- 2) The action of normally-open backup switches is omitted in this article to make the failures tractable.

Based on the above assumption, it can be concluded that the line fault leading to the interruption of downstream load demand and PV output power is equal to the absolute power flow through the corresponding fault line. Hence, (9) can be replaced by the following linear expression:

$$\begin{aligned} MEENS_{t,\eta} &= \sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{nbr}} \sum_{k \in \Omega_{alt}} \lambda_{l,i,k} \delta_{l,i,k} \frac{Dur_{l,i.s.t.\eta}}{365 * 24} P_{l,i.s.t.\eta}^{load.loss} \\ &+ \sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{nbr}} \sum_{k \in \Omega_{alt}} \lambda_{l,i,k} \delta_{l,i,k} \frac{Dur_{l,i.s.t.\eta}}{365 * 24} P_{l,i.s.t.\eta}^{PV.loss} \end{aligned} \quad (37)$$

where $P_{l,i.s.t.\eta}^{load.loss}$ and $P_{l,i.s.t.\eta}^{PV.loss}$ are the curtailed load demand and PV output power due to outage of branch i , respectively.

In (37), the sum of $P_{load,i.s.t.\eta}^{loss}$ and $P_{PV,i.s.t.\eta}^{loss}$ is expressed as the sum of $P_{l,i.s.t.\eta}^{load.loss}$, $P_{l,i.s.t.\eta}^{PV.loss}$, which can be calculated by the following expression:

$$P_{l,i.s.t.\eta}^{load.loss} \geq P_{l,i.s.t.\eta}^{load} - \left(P_{l,i.s.t.\eta}^{PV} + P_{l,i.s.t.\eta}^{DG.cap} \right) \quad (38)$$

$$P_{l,i.s.t.\eta}^{load.loss} \geq 0 \quad (39)$$

$$\begin{aligned} P_{l,i.s.t.\eta}^{load.loss} &\leq P_{l,i.s.t.\eta}^{load} - \left(P_{l,i.s.t.\eta}^{PV} + P_{l,i.s.t.\eta}^{DG.cap} \right) \\ &+ M \left(1 - v_{l,i.s.t.\eta}^{load} \right) \end{aligned} \quad (40)$$

$$P_{l,i.s.t.\eta}^{load.loss} \leq M v_{l,i.s.t.\eta}^{load} \quad (41)$$

$$P_{l,i.s.t.\eta}^{PV.loss} \geq P_{l,i.s.t.\eta}^{PV} - P_{l,i.s.t.\eta}^{load} \quad (42)$$

$$P_{l,i.s.t.\eta}^{PV.loss} \geq 0 \quad (43)$$

$$P_{l,i.s.t.\eta}^{PV.loss} \leq P_{l,i.s.t.\eta}^{PV} - P_{l,i.s.t.\eta}^{load} + M \left(1 - v_{l,i.s.t.\eta}^{PV} \right) \quad (44)$$

$$P_{l,i.s.t.\eta}^{PV.loss} \leq M v_{l,i.s.t.\eta}^{PV} \quad (45)$$

where $P_{l.i.s.t.\eta}^{DG.cap}$ is the sum of installed DG capacity of branch i , $v_{l.i.s.t.\eta}^{load}$ and $v_{l.i.s.t.\eta}^{PV}$ are the auxiliary binary variable.

Validation of the linear expression of curtailed load demand and curtailed PV output power: As can be inferred, if the $P_{l.i.s.t.\eta}^{load}$ is greater than the sum of $P_{l.i.s.t.\eta}^{PV}$ and $P_{l.i.s.t.\eta}^{DG.cap}$, the minimum value of $P_{l.i.s.t.\eta}^{load.loss}$ is set to a positive value by (38), which enforce $v_{l.i.s.t.\eta}^{load}$ to 1 and the (41) is eliminated to make the constraints for $P_{l.i.s.t.\eta}^{load}$ feasible. By contrast, if $P_{l.i.s.t.\eta}^{load}$ is lower than $P_{l.i.s.t.\eta}^{PV}$, the auxiliary variable $v_{l.i.s.t.\eta}^{PV}$ becomes 1 with set $P_{l.i.s.t.\eta}^{PV.loss}$ to $(P_{l.i.s.t.\eta}^{PV} - P_{l.i.s.t.\eta}^{load})$. If the $P_{l.i.s.t.\eta}^{load}$ is greater than $P_{l.i.s.t.\eta}^{PV}$ and lower than the sum of $P_{l.i.s.t.\eta}^{PV}$ and $P_{l.i.s.t.\eta}^{DG.cap}$, all the $v_{l.i.s.t.\eta}^{load}$ and $v_{l.i.s.t.\eta}^{PV}$ becomes 0 with set $P_{l.i.s.t.\eta}^{load.loss}$ and $P_{l.i.s.t.\eta}^{PV.loss}$ to 0.

$P_{l.i.s.t.\eta}^{load}$, $P_{l.i.s.t.\eta}^{PV}$, and $P_{l.i.s.t.\eta}^{DG}$ can be obtained by the following formulation:

$$P_{l.i.s.t.\eta}^{load} = P_{l.i.s.t.\eta}^{load+} + P_{l.i.s.t.\eta}^{load-} \quad (46)$$

$$\sum_{i \in \Omega_{nbr}} \chi_{q,i} (P_{l.i.s.t.\eta}^{load+} - P_{l.i.s.t.\eta}^{load-}) = P_{load,q.s.t.\eta} \quad \forall q \in \Omega_{load} \quad (47)$$

$$\sum_{i \in \Omega_{nbr}} \chi_{q,i} (P_{l.i.s.t.\eta}^{load+} - P_{l.i.s.t.\eta}^{load-}) = 0 \quad \forall q \in \Omega_{Sub} \quad (48)$$

$$(P_{l.i.s.t.\eta}^{load+} - P_{l.i.s.t.\eta}^{load-}) \leq M * Y_{l.i.s.t.\eta}^k \quad (49)$$

$$\frac{1}{M} P_{l.i.s.t.\eta}^{load+} \leq \Lambda_{l.i.s.t.\eta}^{load+} \leq M P_{l.i.s.t.\eta}^{load+} \quad (50)$$

$$\frac{1}{M} P_{l.i.s.t.\eta}^{load-} \leq \Lambda_{l.i.s.t.\eta}^{load-} \leq M P_{l.i.s.t.\eta}^{load-} \quad (51)$$

$$\Lambda_{l.i.s.t.\eta}^{load+} + \Lambda_{l.i.s.t.\eta}^{load-} \leq 1 \quad (52)$$

$$P_{l.i.s.t.\eta}^{PV} = P_{l.i.s.t.\eta}^{PV+} + P_{l.i.s.t.\eta}^{PV-} \quad (53)$$

$$\sum_{i \in \Omega_{nbr}} \chi_{q,i} (P_{l.i.s.t.\eta}^{PV+} - P_{l.i.s.t.\eta}^{PV-}) = P_{PV,q.s.t.\eta} \quad \forall q \in \Omega_{PV} \quad (54)$$

$$\sum_{i \in \Omega_{nbr}} \chi_{q,i} (P_{l.i.s.t.\eta}^{PV+} - P_{l.i.s.t.\eta}^{PV-}) = 0 \quad \forall q \in \Omega_{Sub} \quad (55)$$

$$(P_{l.i.s.t.\eta}^{PV+} - P_{l.i.s.t.\eta}^{PV-}) \leq M * Y_{l.i.s.t.\eta}^k \quad (56)$$

$$\frac{1}{M} P_{l.i.s.t.\eta}^{PV+} \leq \Lambda_{l.i.s.t.\eta}^{PV+} \leq M P_{l.i.s.t.\eta}^{PV+} \quad (57)$$

$$\frac{1}{M} P_{l.i.s.t.\eta}^{PV-} \leq \Lambda_{l.i.s.t.\eta}^{PV-} \leq M P_{l.i.s.t.\eta}^{PV-} \quad (58)$$

$$\Lambda_{l.i.s.t.\eta}^{PV+} + \Lambda_{l.i.s.t.\eta}^{PV-} \leq 1 \quad (59)$$

$$P_{l.i.s.t.\eta}^{DG.cap} = P_{l.i.s.t.\eta}^{DG.cap+} + P_{l.i.s.t.\eta}^{DG.cap-} \quad (60)$$

$$\sum_{i \in \Omega_{nbr}} \chi_{q,i} (P_{l.i.s.t.\eta}^{DG.cap+} - P_{l.i.s.t.\eta}^{DG.cap-}) = Q_{DG,q.s.t.\eta} \quad \forall q \in \Omega_{DG} \quad (61)$$

$$\sum_{i \in \Omega_{nbr}} \chi_{q,i} (P_{l.i.s.t.\eta}^{DG.cap+} - P_{l.i.s.t.\eta}^{DG.cap-}) = 0 \quad \forall q \in \Omega_{Sub} \quad (62)$$

$$(P_{l.i.s.t.\eta}^{DG.cap+} - P_{l.i.s.t.\eta}^{DG.cap-}) \leq M * Y_{l.i.s.t.\eta}^k \quad (63)$$

$$\frac{1}{M} P_{l.i.s.t.\eta}^{DG.cap+} \leq \Lambda_{l.i.s.t.\eta}^{DG.cap+} \leq M P_{l.i.s.t.\eta}^{DG.cap+} \quad (64)$$

$$\frac{1}{M} P_{l.i.s.t.\eta}^{DG.cap-} \leq \Lambda_{l.i.s.t.\eta}^{DG.cap-} \leq M P_{l.i.s.t.\eta}^{DG.cap-} \quad (65)$$

$$\Lambda_{l.i.s.t.\eta}^{DG.cap+} + \Lambda_{l.i.s.t.\eta}^{DG.cap-} \leq 1 \quad (66)$$

where $P_{l.i.s.t.\eta}^{+}$ and $P_{l.i.s.t.\eta}^{-}$ are the power of branch i , when the branch power flow is in the predetermined direction and the opposite one respectively. $\chi_{q,i}$ is the element of node-branch incidence matrix of the distribution network, which is -1 or 1 if the branch i is connected to node q and the flow direction is toward or away from node q . $\Lambda_{l.i.s.t.\eta}^{+}$ and $\Lambda_{l.i.s.t.\eta}^{-}$ are the binary variables to make sure only one of the two positive variable $P_{l.i.s.t.\eta}^{+}$, $P_{l.i.s.t.\eta}^{-}$ be non-zero at time stage t .

Validation of the linear expression of load demand, PV output power, and DG capacity: In these constraints, to avoid negative value of $P_{l.i.s.t.\eta}^{load}$, $P_{l.i.s.t.\eta}^{PV}$, and $P_{l.i.s.t.\eta}^{DG}$, these variables are divided to positive variable as positive and negative part, respectively. (47)–(48), (54)–(55), and (61)–(62) indicate the fictitious nodal balance of load demand, PV output power and DG capacity, respectively. By using (49), (56), and (63), downstream demand, PV output power, and DG capacity of not utilized branches are forced to 0. Additionally, (50)–(52), (57)–(59), and (64)–(66) are utilized to ensure only one of the two positive variable be non-zero at time stage t .

• Linearization of MSAIFI

Under the same assumption as those mentioned for MEENS calculation, the expression of MSAIFI can be rewritten as:

$$MSAIFI_{t,\eta} = \frac{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{nbr}} \sum_{k \in \text{alt}} \lambda_{l,i,k} (n_{l.i.s.t.\eta}^{load} + n_{l.i.s.t.\eta}^{PV})}{\sum_{s \in \Omega_s} \pi_s \sum_{i \in \Omega_{nbr}} N_{load,i.s.t.\eta}} \quad (67)$$

where $n_{l.i.s.t.\eta}^{load}$ and $n_{l.i.s.t.\eta}^{PV}$ are the number of power customers and PV associated with failure of line i , respectively.

Based on these auxiliary variables, the following equations can be derived:

$$n_{l.i.s.t.\eta}^{load} = n_{l.i.s.t.\eta}^{load+} + n_{l.i.s.t.\eta}^{load-} \quad (68)$$

$$\sum_{i \in \Omega_{nbr}} \chi_{q,i} (n_{l.i.s.t.\eta}^{load+} - n_{l.i.s.t.\eta}^{load-}) = n_{load,q.s.t.\eta} \quad \forall q \in \Omega_{load} \quad (69)$$

$$\sum_{i \in \Omega_{nbr}} \chi_{q,i} (n_{l.i.s.t.\eta}^{load+} - n_{l.i.s.t.\eta}^{load-}) = 0 \quad \forall q \in \Omega_{sub} \quad (70)$$

$$(n_{l.i.s.t.\eta}^{load+} + n_{l.i.s.t.\eta}^{load-}) \leq M * Y_{l.i.s.t.\eta}^k \quad (71)$$

$$\frac{1}{M} n_{l.i.s.t.\eta}^{load+} \leq \Lambda_{l.i.s.t.\eta}^{load+} \leq M n_{l.i.s.t.\eta}^{load+} \quad (72)$$

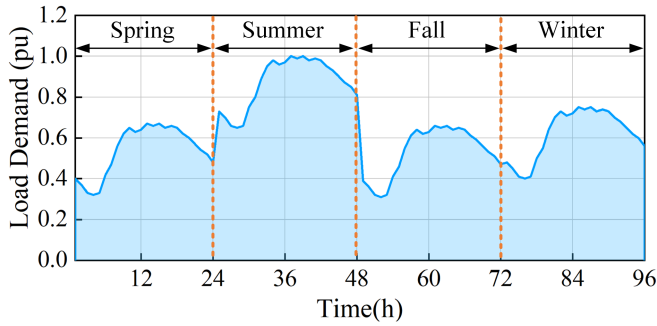


Fig. 6. Load demand for a typical day in each scenario.

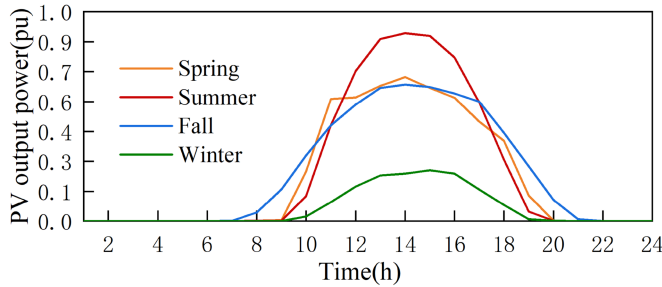


Fig. 7. PV output power for a typical day in each scenario.

TABLE I
CANDIDATE INVESTMENT CHOICE OF REPLACING AND ADDING BRANCHES

ALTERNATIVE	Option1	Option2	Option3
Capacity(MVA)	3.89	6.17	3.98
Investment Cost(\$10 ³ /km)	14.87	24.98	18.1
Failure Rate(failure/km/year)	0.3	0.3	0.2
Repair Time(hours/repair)	0.5	0.5	0.5

TABLE II
CANDIDATE NODE FOR ADDING DG

Node	Capacity(MW)	Investment Cost(\$10 ⁴)
	0.5	25
1-20	1	40
	2	60

based on real historical data, as shown in Fig. 7. In this case, the operation strategies based on the operation cost and operation reliability cost in each load and PV scenario are co-optimized with the investment decisions in the proposed planning model. The details of various alternative construction investment decisions for replacing new feeders, adding new feeders, and applying DGs are shown in Tables I–II. The discount rate of interest i is 10.

Four different cases are considered to better investigate the effectiveness of the proposed multi-stage planning method considering the reliability index and microgrid operation for distribution networks with high penetration renewable energy. The simulation cases are listed below:

TABLE III
SIMULATION RESULT FOR THE 24-NODE SYSTEM

Case	Case A	Case B	Case C	Case D
Conditions				
DG	×	✓	✓	✓
TC	×	×	✓	✓
Microgrid	×	×	×	✓
Investment	21.69	17.46	17.71	16.32
SC	13.71	14.06	14.04	13.89
Cost(\$10 ⁶)				
TC	4.376	4.09	3.04	2.67
Total	39.775	35.4	34.79	32.88

Case A: Distribution planning without the integration of DGs, and the inclusion of microgrid, without incorporating operation reliability cost.

Case B: Distribution planning with the integration of DGs, without the inclusion of microgrid, without incorporating operation reliability cost.

Case C: Distribution network planning with the integration of DGs and incorporating operation reliability cost without the inclusion of microgrid.

Case D: Distribution network planning with the integration of DGs, incorporating operation reliability cost and the inclusion of microgrid.

A. Planning Result and Analysis

Outcomes associated with these cases are shown in Table III. In Case A, without the integration of DGs, and the inclusion of microgrid, without incorporating operation reliability cost, the total cost is $\$39.755 \times 10^6$, which are both much higher than those of other cases, given the load demand can only be solved by adding high capacity feeder and substation. Compared Case A with the other three cases, it can be inferred that the installation of DG units can significantly decrease the total planning cost by reducing the total cost associated with the substation and branch. Compared Case B with Case C, the former where the operation reliability cost is eliminated in the objective function leads to a higher total cost. The total cost of Case C is lower in Case D when the microgrid operation is taken into consideration, which provides flexible topology during the planning horizon to further decrease the operation reliability cost.

The planning results of all cases are depicted in Figs. 8 and 9, where continuous lines are circuits in operation, and the load node with a red circle means that DGs are added in this planning stage. It can be observed from Fig. 8 that the planning result of Case A is altered with the integration of DGs in Case B. At stage 3, with the continuous growth of load demand in node 1, the supply power at node 14 was changed from substation 21 to substation 24 to limit the voltage drop due to line loss in Case A, which can be solved by adding DG in Case B. Fig. 8 reflects the benefits of the planning model with the integration of DGs, which can improve the nodal voltages by reducing branches power flow with the growth of load demand, leading to a significant reduction in the investment cost.

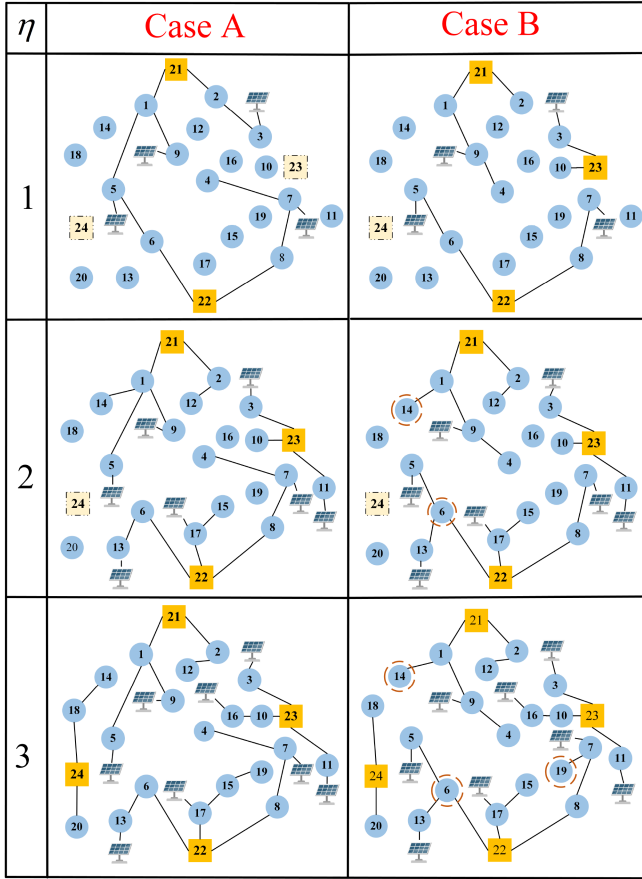


Fig. 8. Planning results of Case A and Case B.

A glance at the planning result for Case B and Case C demonstrated that in the latter case, more DGs are added in load nodes to deal with the load demand and the uncertainty of PV in the distribution network. This is because the planning problem in Case C is sensitive to the network reliability level after incorporating operation reliability cost in the objective function. Even though these result in higher investment costs resulting from more DG capacity, the operation reliability cost is remarkably decreased, and the system reliability is enhanced in Case C.

Take the optimized topology of Cases C and Case D as an example. Although the final planning result in Case D is almost identical to that in Case C, the load at node 4 is only supplied by DG in Stage 1, and the construction of branches 4–9 is delayed. Similar situations occur at node 5 in stage 2 (node 5 was operated at microgrid operation when the load demand is relatively low to reduce the penalty cost of operational reliability with the line failure of branch 5–6), which reduce the investment costs and operation reliability cost.

To further verify the effectiveness of the proposed MILP model, the solving result of different planning models is compared, as shown in Table IV, where MINLP is the original model and MILP represents the proposed linearization model. All the model has almost the same solution result, and the MINLP is solved in more than 60 h due to the nonlinear expressions of operation reliability cost, while it only takes 0.8 h when

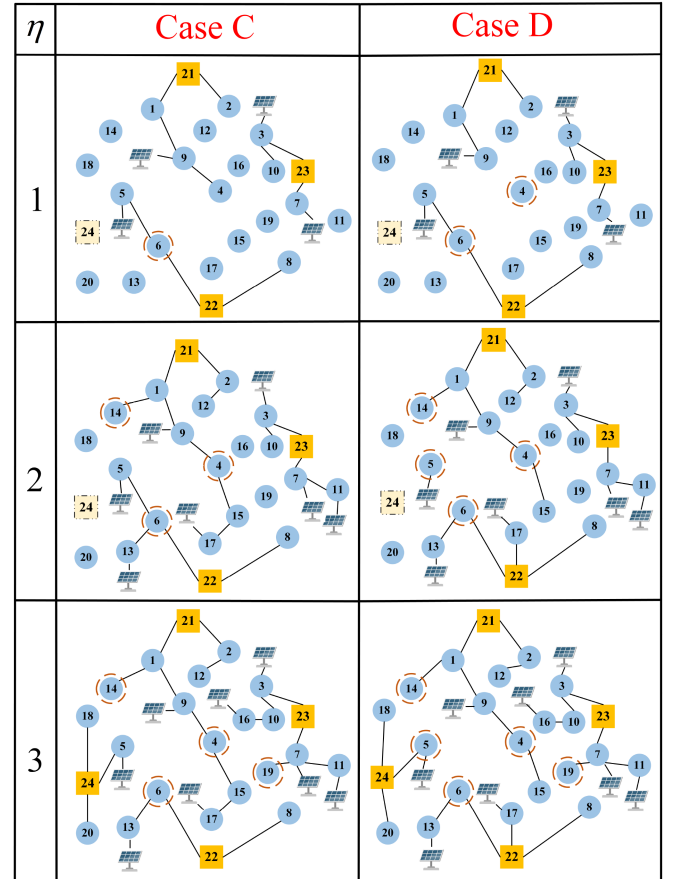


Fig. 9. Planning results of Case C and Case D.

TABLE IV
SOLVING RESULT OF DIFFERENT PLANNING MODEL

Model	Total Cost(\$10 ⁶)	Solution Method	Solution Time
MINLP	32.85	Genetic Algorithm	> 60h
MILP	32.88	GUROBI	0.8h

employing the MILP model. The results demonstrate that the linearization method has a better performance in execution time and accuracy for the planning result.

B. Simulation Result for the 54-Node System

The 54-node test system which is described in [30] is utilized in this article to further investigate the effectiveness of the proposed planning model. This test system is comprised of 50 load points, 4 substations, and 63 branches including 9 fixed, and 8 reinforceable feeder sections, as well as 46 candidate branches for adding to the network.

A summary of the simulation result for the 54-node system is presented in Table V. A glance at this table reveals that the reduction of the operation reliability cost in the proposed model is dramatically higher than that in the 24-node system due to more flexible topology choice. The operation reliability cost is $10.27 \times \$10^6$ and $9.92 \times \$10^6$ in Case A and Case B, respectively, which are decreased to $6.04 \times \$10^6$ in Case C

TABLE V
SIMULATION RESULT FOR THE 54-NODE SYSTEM

Case	Case A	Case B	Case C	Case D
Conditions				
DG	×	✓	✓	✓
TC	×	×	✓	✓
Microgrid	×	×	×	✓
Investment	26.89	21.76	21.81	20.13
SC	19.47	19.45	19.41	18.81
Cost(\$10 ⁶)				
TC	10.27	9.92	6.04	3.98
Total	56.63	51.13	47.24	42.92

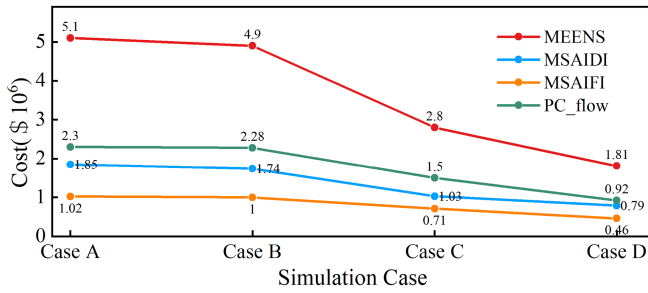


Fig. 10. Operation Reliability Cost for The 54-Node System.

with the addition of modified reliability index formulation in the objective function. The operation reliability cost is further reduced to $3.98 \times \$10^6$ in Case D when the microgrid mode is allowed in the operation strategy.

Modified reliability index of the 54-node test system over the whole planning horizon (15 years) are depicted in Fig. 10. It can be observed that all the reliability index (MENSS, MSAIDI, and MSAIFI) perform the same trend in the various simulation cases. The results also reveal that MSAIDI and MSAIFI are relatively insignificant for the system operator compared with MEENS in the planning decision.

In summary, the model takes into account investment cost and operation cost of substation, branch, and DG, as well as the operation reliability cost containing the penalty cost incurred of undelivered energy and PV power loss and the negative impact of PV on system operation reliability. Additionally, the topology constraint considering microgrid is applied in the proposed planning model to improve the system operation reliability. Simulation results of various cases prove the effectiveness of the proposed planning method. The benefits of the proposed ADN planning are listed as follows:

1. It offers the possibility of improving the operation reliability of the system with higher penetration level PV according to the power flow control;
2. It decrease the load demand loss and PV output power loss and the negative impact of PV on system operation reliability by imposing the operation reliability cost in the proposed planning model.
3. It introduces the microgrid operation which can delay the DN branch alterations and reduce the operation reliability cost of distribution network.

4. It can be solved easily by some commercial solvers such as CPLEX and GUROBI through the linear expression of constraints.

VI. DISCUSSION

A. Effectiveness of Distribution Network Reconfiguration Including Microgrid

In this article, the distribution network reconfiguration including microgrid is considered at the operation level to improve system operation reliability, which is realized by the proposed topology constraints shown in (16)–(36). Effectiveness of distribution network reconfiguration including microgrid are presented in Figs. 8 and 9.

It can be observed that the system topology in Case A, Case B, and Case C always consist of radial trees with a substation being root node due to strictly radial constraints. In Case D, where separated load node with DGs is allowed if microgrid operation is allowed in the operation strategy, some nodes are supplied solely by DG without construction of branch. The flexible topology constraints introduce additional options for investment planning, which can delay the system branch alterations, reduce the total cost and the operation reliability cost of distribution network.

B. Advantages and Applicability of Proposed Planning Model

As numerical results illustrate, the proposed planning model with the integration of DGs, incorporating operation reliability cost and the inclusion of microgrid is able to maximize the benefits of the distribution network with high penetration renewable energy by decreasing the planning cost (investment cost + operation cost) and improving the system operation reliability.

The scenarios-based optimization approach is utilized in this article to deal with the uncertainty of PV installed in the distribution network. Although all the PV operations in the planning stage are divided into four scenarios based on the seasonal power, the concept of PV scenario provided in this article is changeable depending on the real planning requirements. Only the integration of DGs is considered in the proposed planning model. Consideration of the energy storage system (ESS) could provide more effective planning strategies for the distribution network with high penetration renewable energy, which can be taken as future work.

VII. CONCLUSION

This article proposes multi-stage planning model for the distribution network with high penetration PV, where all the investment decision ($X_{SUB,i}$, $X_{line,i}$) and operation decision ($Y_{line,i}$, $P_{DG,i}$, $P_{SUB,i}$) are co-optimized together in the whole planning horizon. The model takes into account investment cost and operation cost of substation, branch, and DG, as well as the operation reliability cost containing the penalty cost incurred of undelivered energy and PV power loss and the negative impact of PV on system operation reliability. Additionally, the topology constraint considering microgrid is applied in the proposed planning model to improve the system operation reliability.

To address the computational burden, a series of linearization methods is utilized to obtain a MILP formulation for the planning problem. The use of an MILP formulation has the following benefits: (a) an efficient computational behavior with existing MILP solvers; and (b) guarantees convergence to optimality using classical optimization techniques.

Simulation results of various cases prove the effectiveness of the proposed planning method. The advantage of the integration of DGs are discussed first, then the benefit of incorporating operation reliability cost in the objective function is shown by comparing Case B and Case C, which encourage the system operator to add more DG and adjust network topology for high operation reliability levels. Finally, the topology constraint allowing microgrid in operation maximize the system operation reliability due to flexible topology. It also offers a new perspective planning method for the distribution network with high penetration renewable energy.

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